Question 1 - Records of all experiments

Question 1: Record all commands, pipeline configurations, screenshots, commit history, and observations for each experiment.

Repository: https://github.com/ksamirdev/ensp461-devops-lab1

Commit history

```
* 81a3761 2026-11-05 18:29 Add lab command logs, screenshots and tooling
* e93d90f 2026-11-05 17:41 Bump release to 1.1.0 to exercise automatic deployment
* 8633054 2026-11-05 17:39 Add declarative Jenkins pipeline with build, test, package and deploy stages
* a34c4f5 2026-11-05 17:39 Add systemd unit and health-checked deploy script
*   3d80611 2026-11-05 17:35 Merge feature/test-suite into main
|\
| * 9c2335f 2026-11-05 17:35 Add pytest suite for service endpoints
|/
* 7b5d99f 2026-11-05 17:35 Add text statistics Flask service
```

Git installation and configuration (Q4)

```
$ cat /etc/os-release | head -2
PRETTY_NAME="Ubuntu 24.04.5 LTS"
NAME="Ubuntu"

$ sudo apt-get update -qq

$ sudo apt-get install -y -qq git

$ git --version
git version 2.43.0

$ git config --global user.name 'Samir Kumar Gupta'

$ git config --global user.email 'samirdiff@proton.me'

$ git config --global init.defaultBranch main

$ git config --global --list
user.name=Samir Kumar Gupta
user.email=samirdiff@proton.me
init.defaultbranch=main
```

Fig: Git setup

Jenkins installation (Q4)

```
$ curl -fsSL https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key | sudo tee
/usr/share/keyrings/jenkins-keyring.asc >/dev/null

$ echo 'deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc] https://pkg.jenkins.io/debian-stable
binary/' | sudo tee /etc/apt/sources.list.d/jenkins.list
deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc] https://pkg.jenkins.io/debian-stable binary/

$ sudo apt-get update -qq

$ sudo apt-get install -y -qq jenkins
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package net-tools.
Preparing to unpack .../net-tools_2.10-0.1ubuntu4.4_arm64.deb ...
Unpacking net-tools (2.10-0.1ubuntu4.4) ...
Selecting previously unselected package jenkins.
Preparing to unpack .../jenkins_2.580.2_all.deb ...
Unpacking jenkins (2.580.2) ...
Setting up net-tools (2.10-0.1ubuntu4.4) ...
Setting up jenkins (2.580.2) ...
invoke-rc.d: policy-rc.d denied execution of start.
Created symlink /etc/systemd/system/multi-user.target.wants/jenkins.service →
/usr/lib/systemd/system/jenkins.service.

/usr/sbin/policy-rc.d returned 101, not running 'start jenkins.service'

$ systemctl is-active jenkins
inactive

$ dpkg -s jenkins | grep -E '^(Package|Version)'
Package: jenkins
Version: 2.580.2

$ sudo ss -ltnp | grep 8080
```

Fig: Jenkins install

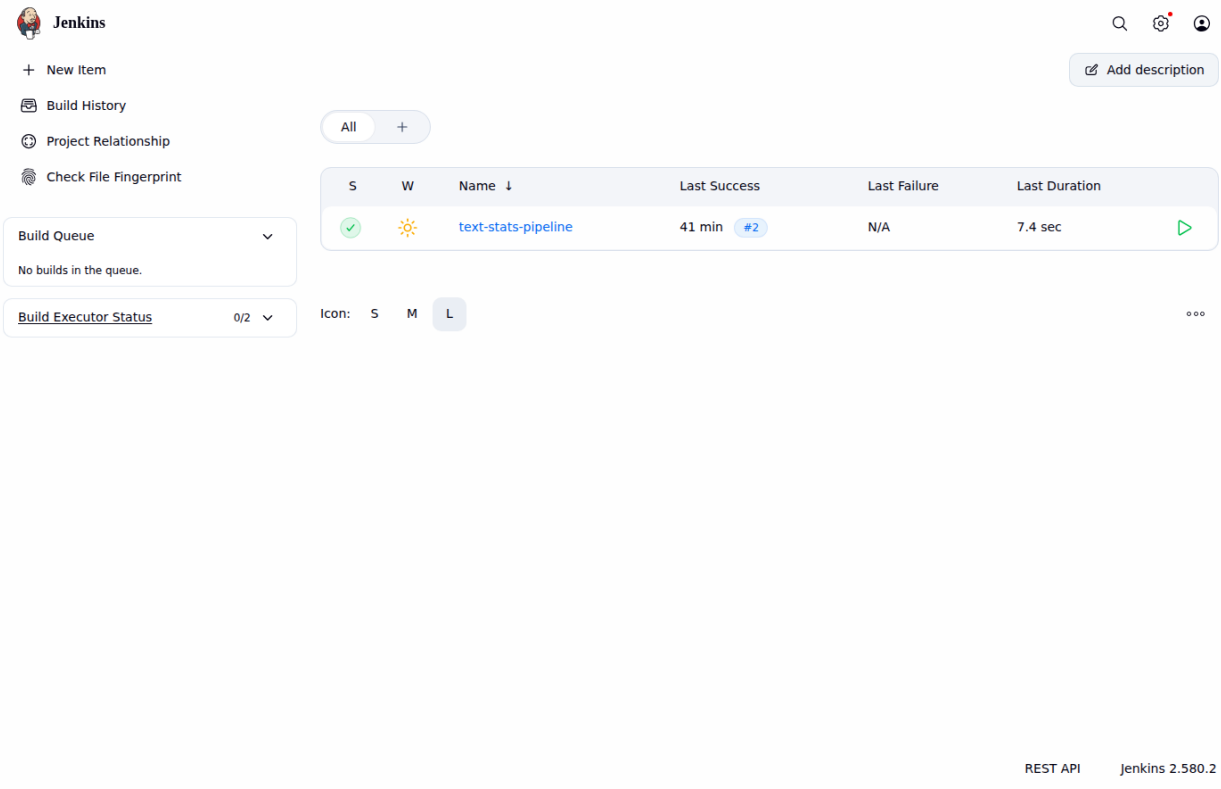


Fig: Dashboard

Version control workflow (Q5)

```
$ git init
Initialized empty Git repository in /home/samir/ensp461-devops-lab1/.git/

$ git status --short
?? .gitignore
?? README.md
?? requirements.txt
?? service/

$ git add .

$ git commit -m 'Add text statistics Flask service'
[main (root-commit) 7b5d99f] Add text statistics Flask service
 5 files changed, 63 insertions(+)
 create mode 100644 .gitignore
 create mode 100644 README.md
 create mode 100644 requirements.txt
 create mode 100644 service/__init__.py
 create mode 100644 service/web.py

$ git checkout -b feature/test-suite
Switched to a new branch 'feature/test-suite'

$ git add tests

$ git commit -m 'Add pytest suite for service endpoints'
[feature/test-suite 9c2335f] Add pytest suite for service endpoints
 1 file changed, 29 insertions(+)
 create mode 100644 tests/test_web.py

$ python3 -m venv .venv && .venv/bin/pip install -q -r requirements.txt && .venv/bin/python -m pytest -v
============================= test session starts ==============================
platform linux -- Python 3.12.3, pytest-8.3.4, pluggy-1.7.0 -- /home/samir/ensp461-devops-
lab1/.venv/bin/python
cachedir: .pytest_cache
rootdir: /home/samir/ensp461-devops-lab1
collecting ... collected 5 items

tests/test_web.py::test_healthz_reports_healthy PASSED                   [ 20%]
tests/test_web.py::test_text_stats_counts_unique_words_case_insensitively PASSED [ 40%]
tests/test_web.py::test_stats_endpoint_returns_counts PASSED             [ 60%]
tests/test_web.py::test_stats_endpoint_rejects_invalid_text[body0] PASSED [ 80%]
tests/test_web.py::test_stats_endpoint_rejects_invalid_text[body1] PASSED [100%]

============================== 5 passed in 0.05s ===============================

$ git checkout main
Switched to branch 'main'

$ git merge --no-ff feature/test-suite -m 'Merge feature/test-suite into main'
Merge made by the 'ort' strategy.
 tests/test_web.py | 29 ++++++++++++++++++++++++++++
 1 file changed, 29 insertions(+)
 create mode 100644 tests/test_web.py

$ git tag -a v1.0.0 -m 'First tested release'

$ git branch -a
  feature/test-suite
* main
```

Fig: Git workflow

```
$ git push origin main --tags
To https://github.com/ksamirdev/ensp461-devops-lab1.git
   3d80611..8633054  main -> main

$ git push origin main --tags
To https://github.com/ksamirdev/ensp461-devops-lab1.git
   8633054..e93d90f  main -> main
 * [new tag]         v1.1.0 -> v1.1.0
```

Fig: Pushes

Pipeline job and execution (Q6, Q7)

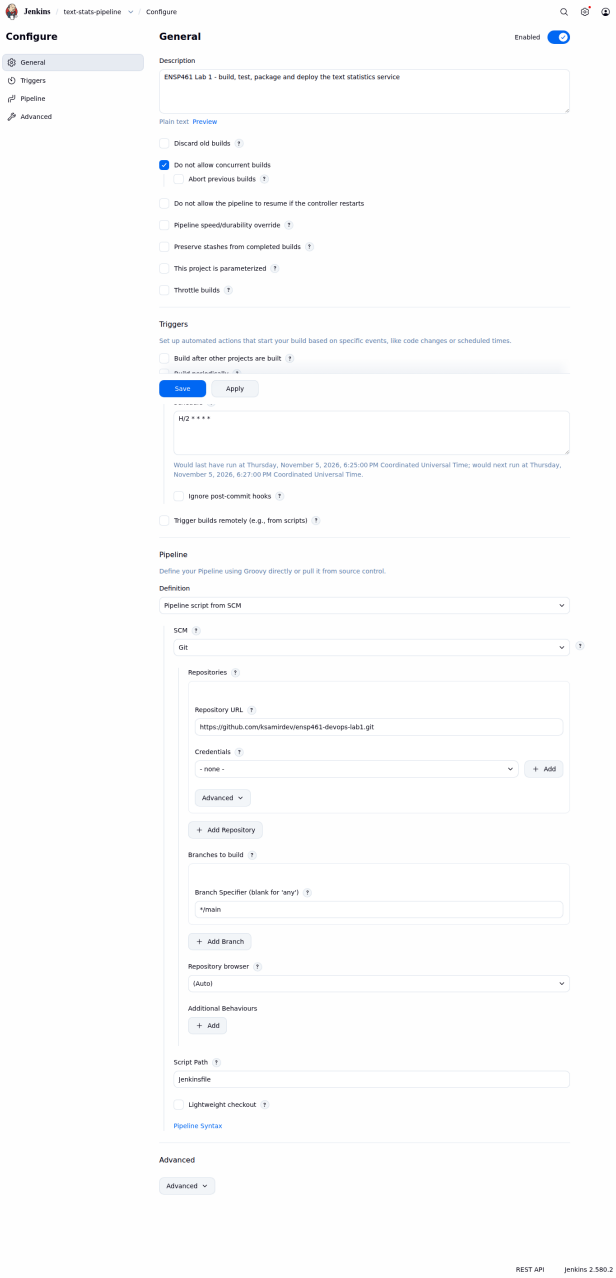


Fig: Job configuration

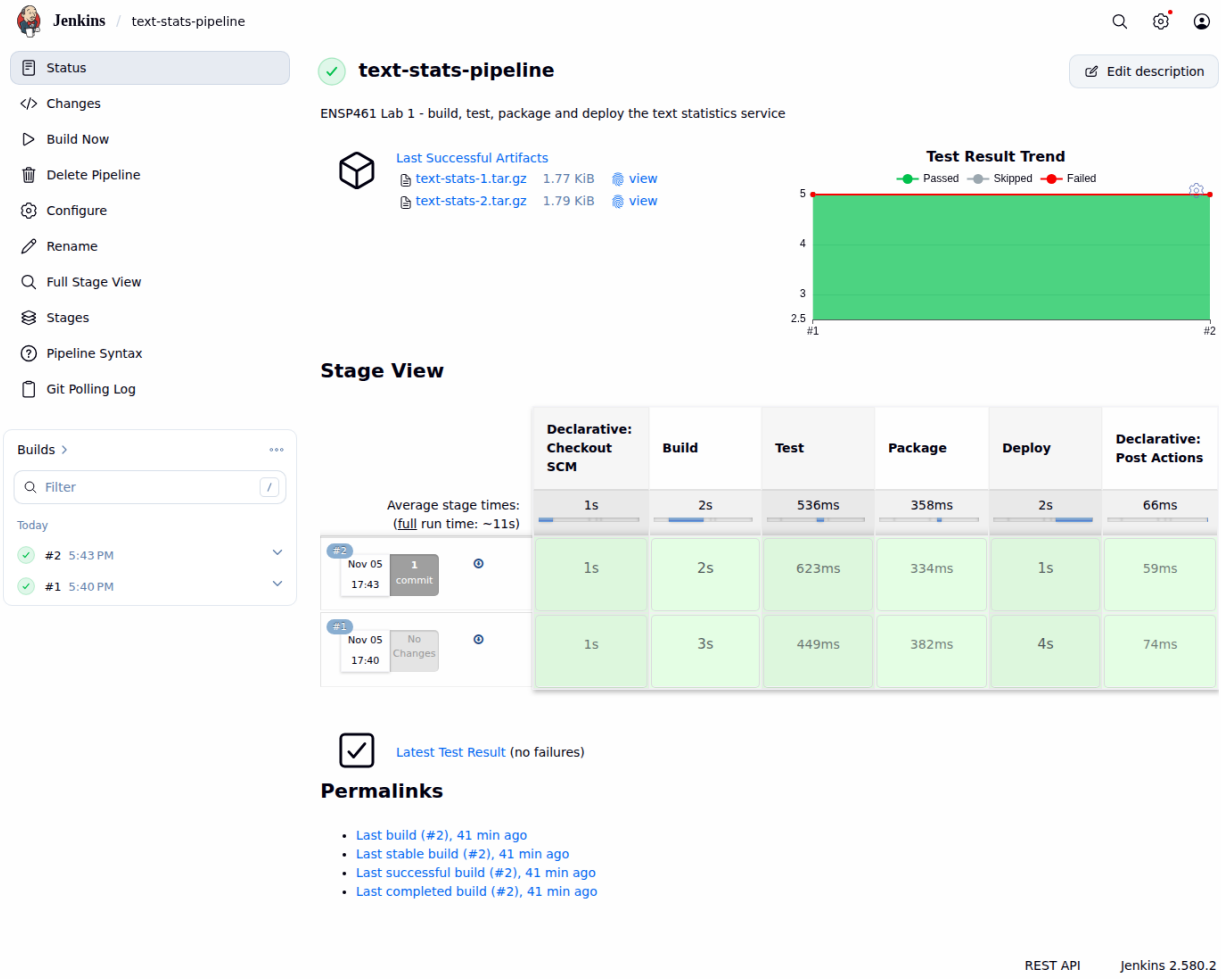


Fig: Stage view

```
Started by user Samir Kumar Gupta
Checking out git https://github.com/ksamirdev/ensp461-devops-lab1.git into /var/lib/jenkins/workspace/text-
stats-pipeline@script/c7d6e329e59a10382091761ebc2c41deb57551efdc0ed28a2146fb72bc04d9f1 to read Jenkinsfile
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/ksamirdev/ensp461-devops-lab1.git
Fetching upstream changes from https://github.com/ksamirdev/ensp461-devops-lab1.git
Avoid second fetch
Checking out Revision 86330543fd993df3600eda68a0140ce14d952143 (refs/remotes/origin/main)
Commit message: "Add declarative Jenkins pipeline with build, test, package and deploy stages"
First time build. Skipping changelog.
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in /var/lib/jenkins/workspace/text-stats-pipeline
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: Checkout SCM)
[Pipeline] checkout
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/ksamirdev/ensp461-devops-lab1.git
Fetching upstream changes from https://github.com/ksamirdev/ensp461-devops-lab1.git
Avoid second fetch
Checking out Revision 86330543fd993df3600eda68a0140ce14d952143 (refs/remotes/origin/main)
Commit message: "Add declarative Jenkins pipeline with build, test, package and deploy stages"
[Pipeline] withEnv
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Build)
[Pipeline] sh
+ python3 -m venv .venv
[Pipeline] sh
+ .venv/bin/pip install -q -r requirements.txt
[Pipeline] stage
[Pipeline] { (Test)
[Pipeline] sh
+ .venv/bin/python -m pytest -v --junitxml=test-results.xml
============================= test session starts ==============================
platform linux -- Python 3.12.3, pytest-8.3.4, pluggy-1.7.0 -- /var/lib/jenkins/workspace/text-stats-
pipeline/.venv/bin/python
cachedir: .pytest_cache
rootdir: /var/lib/jenkins/workspace/text-stats-pipeline
collecting ... collected 5 items

tests/test_web.py::test_healthz_reports_healthy PASSED                   [ 20%]
tests/test_web.py::test_text_stats_counts_unique_words_case_insensitively PASSED [ 40%]
tests/test_web.py::test_stats_endpoint_returns_counts PASSED             [ 60%]
tests/test_web.py::test_stats_endpoint_rejects_invalid_text[body0] PASSED [ 80%]
tests/test_web.py::test_stats_endpoint_rejects_invalid_text[body1] PASSED [100%]

- generated xml file: /var/lib/jenkins/workspace/text-stats-pipeline/test-results.xml -
============================== 5 passed in 0.05s ===============================
Post stage
[Pipeline] junit
Recording test results
[Checks API] No suitable checks publisher found.
[Pipeline] stage
[Pipeline] { (Package)
```

Fig: Build 1 console

Automated deployment (Q3)

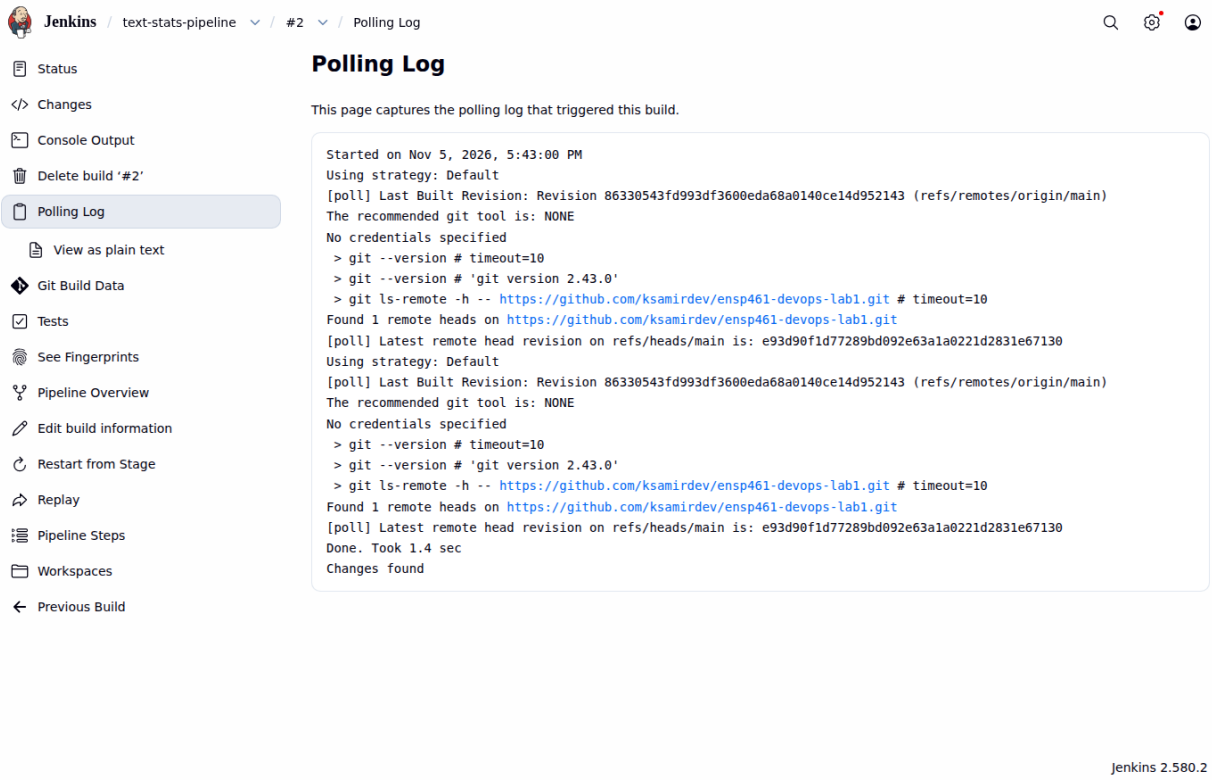


Fig: Polling log

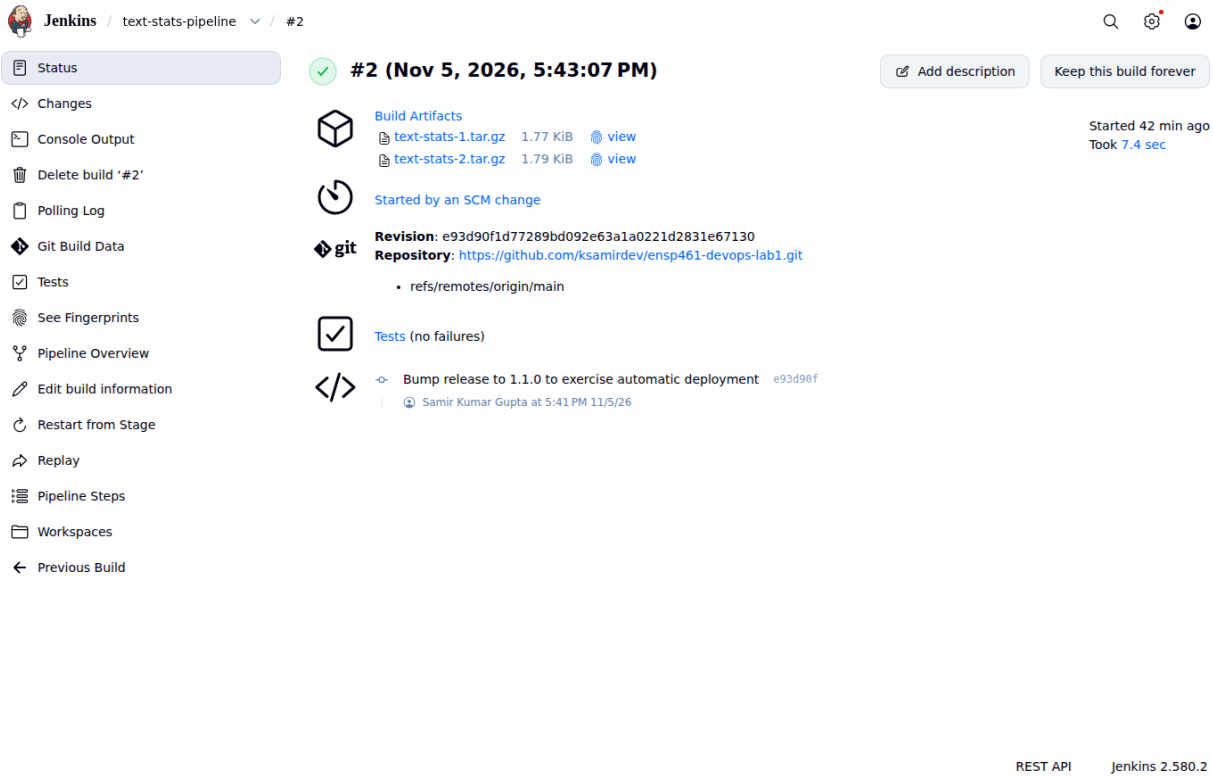


Fig: Build 2

```
$ systemctl is-active text-stats
active

$ curl -s http://localhost:5000/healthz
{"healthy":true,"release":"1.0.0"}

$ curl -s http://localhost:5000/
{"release":"1.0.0","service":"text-stats"}

$ curl -s -X POST -H 'Content-Type: application/json' -d '{"text":"build test build"}'
http://localhost:5000/stats
{"characters":16,"unique_words":2,"words":3}

$ curl -s -o /dev/null -w '%{http_code}\n' -X POST -H 'Content-Type: application/json' -d '{}'
http://localhost:5000/stats
400

$ curl -s http://localhost:5000/healthz
{"healthy":true,"release":"1.1.0"}

$ ls -l /opt/text-stats
total 4
-rw-r--r-- 1 jenkins jenkins 27 Nov  5 17:40 requirements.txt
drwxr-xr-x 1 jenkins jenkins 56 Nov  5 17:43 service
```

Fig: Service checks

Pipeline configuration

```
pipeline {
    agent any
    options { disableConcurrentBuilds() }
    triggers { pollSCM('H/2 * * * *') }

    stages {
        stage('Build') {
            steps {
                sh 'python3 -m venv .venv'
                sh '.venv/bin/pip install -q -r requirements.txt'
            }
        }
        stage('Test') {
            steps {
                sh '.venv/bin/python -m pytest -v --junitxml=test-results.xml'
            }
            post {
                always { junit 'test-results.xml' }
            }
        }
        stage('Package') {
            steps {
                sh 'mkdir -p dist && tar czf dist/text-stats-${BUILD_NUMBER}.tar.gz service requirements.txt'
                archiveArtifacts artifacts: 'dist/*.tar.gz', fingerprint: true
            }
        }
        stage('Deploy') {
            steps {
                sh 'sudo /usr/local/bin/text-stats-deploy "$WORKSPACE/dist/text-stats-${BUILD_NUMBER}.tar.gz"'
            }
        }
    }

    post {
        success { echo "Build ${BUILD_NUMBER} deployed" }
        failure { echo "Build ${BUILD_NUMBER} failed, deployed release unchanged" }
    }
}
```

Observations

- Git and Jenkins both installed from apt; the container's minimal image also needed python3-venv, curl and fontconfig.
- Global identity set with git config lives in ~/.gitconfig and applies to every repository for that user.
- Jenkins 2.580.2 was installed with OpenJDK 21, using the 2026 Jenkins repository signing key.
- The apt post-install step did not start Jenkins inside the container (policy-rc.d denied it), so the service was started with systemctl.
- The feature branch kept main unchanged until the test suite passed.
- Merging with --no-ff keeps a merge commit, so the graph shows when the feature landed.
- An annotated tag marks the first tested release, v1.0.0.
- Output of the very first push was not captured. The later push log and git ls-remote output confirm the branch and tag on GitHub.
- Build 1 was started manually and finished with SUCCESS in about 15 seconds.
- Pipeline script from SCM means the pipeline definition is versioned with the code it builds.
- Using a public HTTPS remote means Jenkins needs no stored Git credentials.
- Declarative stages make failures easy to locate because each stage is reported separately.
- The junit step publishes test results even when tests fail because it runs in post { always }.
- In build 1 the first health probe failed while the service was starting, and the retry loop in the deploy script passed on a later attempt.
- Build 2 was started by an SCM change, without clicking Build Now, and finished in about 7 seconds.
- After build 2 the /healthz endpoint reported release 1.1.0, confirming the new code went live.
- sudo for the jenkins user is limited to one root-owned script. That narrows the rule but is not a full security boundary, because the script still installs Python packages from the archive as root.
- One later poll took 58 seconds and logged a stack trace, with no effect on either build. The cause was not investigated, and later polls completed normally.